MILITARY SPECIFICATION

WHEEL, ABRASIVE, SILICON CARBIDE, DENTAL MODEL TRIMMING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers an abrasive wheel for use in the dental prosthetic laboratory.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-Up, Paperboard.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof Or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

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DRAWINGS

Armed Services Medical Procurement Agency (ASMPA)

10187 Wheel, Abrasive, Dental
Model Trimming, 3 by 5/8 inches

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 OTHER PUBLICATIONS. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNITED STATES DEPARTMENT OF COMMERCE - NATIONAL BUREAU OF STANDARDS

Commercial Standard CS 271-65.
Grading of Abrasive Grain for Grinding Wheels

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402).

3. REQUIREMENTS

3.1 Materials.

3.1.1 Abrasive. The abrasive shall be high quality silicon carbide crystals, free from extraneous material and suitable for the purpose intended. The size of the abrasive grain shall be grit No. 20, in conformance with table I of Commercial Standard CS 271-65.

3.1.2 Binder. The binder for the abrasive shall be a suitable nonvitrifying ceramic binder.

3.2 Style, design and dimensions. The wheels shall be suitable for use with dental laboratory grinding and polishing machines. The wheels shall have a square edge and shall be suitable for use in the dental prosthetic laboratory for trimming plaster and artificial stone models. The style, design and dimensions of the wheels shall conform to Armed Services Medical Procurement Agency (ASMPA) Drawing No. 10187.

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3.3 Construction. The silicon carbide abrasive wheel shall be built around a hub assembly made of lead, brass, and brass wire mesh. Lead shall form a press fit with the brass rings. The hub assembly shall be flush with the ceramic part of the wheel on both sides. The wheel shall be symmetrically made, and the hole shall be centered with respect to the wheel diameter. Side faces of wheel shall be flat with respect to each other.

3.4 Performance. The wheel shall run true and wear evenly, and the binder shall not fuse, glaze, or create heat when in use, when tested as specified in 4.4.1.

3.5 Workmanship. Workmanship shall be first class throughout. Wheels shall be free from defects which detract from their appearance or impair their serviceability, proper functioning, or intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the components or materials, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component or material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered are classified as specified in Tables I and III (see 4.2.1 and 4.3) and Table II (see 4.2.2).

4.2.1 Sampling for examination. A random sample of wheels shall be selected from each lot in accordance with MIL-STD-105. The inspection level and Acceptable Quality Level (AQL) shall be as indicated in Table I. Unit of product for examination purposes shall be one wheel.

Table I. Examination Criteria

Type defect	Inspection level	AQL (Percent defective)
Major	II	2.5
Minor	S-2	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II, except that the acceptance number shall be zero for all sample sizes. The unit of product for test purposes shall be one wheel.

Table II. Test Criteria

Characteristic	Requirement	Test procedure	Inspection level
Performance	3.4	4.4.1	S-2

4.3 Examination. The wheels shall be examined to determine compliance with the requirements contained in this specification. Examination shall be conducted as specified in 4.2.1. Any wheel in the sample containing one or more defects shall be rejected, and if the number of defective wheels in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. The wheels shall be examined for defects including, but not limited to, those listed in Table III.

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Table III. Classification of defects

Categories	Defects
Major:	
101	Style and design of wheel not as specified.
102	Construction not as specified.
103	Dimensions of center hole not as specified.
Minor	
201	Packaging and packing not as specified.
202	Marking on packaging and pack not as specified.
203	Dimensions (other than 103) not as specified.

4.4 Test procedure.

4.4.1 Performance. A practical performance test shall be conducted to determine compliance with the requirements in 3.4.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (6.1).

5.1.1 Unit of issue. One wheel, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	288 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

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5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors shall be acceptable. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors shall be acceptable, or PPP-B-636. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof in accordance with paragraph 30.4 of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Limitations. This specification does not cover all types, classes, grades, styles or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

6.3 National Stock Number (NSN) coverage. This specification covers the following item appearing in the Federal Supply Catalog:

Stock Number	Item Identification
6520-00-539-1300	WHEEL, ABRASIVE, Silicon Carbide, Dental Model Trimming, Square Edge, 3 by 5/8 inches

6.4 Supersession data. This specification includes the requirements of Purchase Description No. 5, dated 16 September 1963.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6520-1364

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-W-37013

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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